

This is sample data provided for CALIBER purposes only		ONE POINT LESSON (OPL)		OPL No: OPL-DC-3401A-07 Sheet 1 of 1
OPL Title	Manway Gasket & Box-Up After Catalyst Change	Discipline	Mechanical	
Equipment	DC-3401A - CATALYST REDUCTION REACTOR	Area / Unit	3400 - CATALYST REDUCTION SYSTEM	
Related Interlock	SEQ-3401 (SIL 2)	P&ID Ref	TJC-LLD-PID-3401	
Classification:	☐ Basic Knowledge	☒ Improvement	☐ Trouble Case	Aspect: P-Q-C-D-S-M-E

1. PURPOSE / OBJECTIVE

This One Point Lesson explains **manway gasket & box-up after catalyst change** for DC-3401A (CATALYST REDUCTION REACTOR) in the CATALYST REDUCTION SYSTEM. A leaking manway loses inert gas and risks air ingress that could ignite the fresh, active catalyst. The lesson gives operators and technicians the key knowledge and the correct step-by-step method so the task is done safely, correctly and consistently every time.

2. SAFETY PRECAUTIONS

Hazard note: this equipment is **HIGH CRITICAL** handling **N2 / H2 / Pd catalyst** at up to 6 barg / FV / 250 degC. Observe the following before and during the task:

- Apply LOTO (Lock-Out Tag-Out) and prove zero energy before starting work.
- Isolate, depressurise, drain and purge the equipment; obtain a valid work permit.
- Wear the correct PPE: coverall, safety shoes, gloves, goggles and hearing protection.
- Beware of stored mechanical energy (springs, rotating inertia) and pinch points.
- Use certified lifting gear and correct manual-handling technique for heavy parts.

3. TOOLS & MATERIALS REQUIRED

- Standard mechanical hand tools & calibrated torque wrench
- LOTO padlocks and permit
- Laser/dial alignment kit (as applicable)
- Feeler gauges, dial indicator, seal/bearing pullers
- Correct spare parts and gaskets per the BOM
- Lifting gear and slings (certified)

4. DETAILED PROCEDURE / STEPS

Step	Action	Check / Acceptance
1	Always fit a new spiral-wound SS316/graphite gasket	Isolation & LOTO verified
2	Torque the manway bolts in a star pattern to the specified value	Parts match BOM / spec
3	Leak test the joint with nitrogen at 2 kg/cm2	Torque / clearance to spec
4	Record the catalyst batch number and the loaded volume (3.2 m3)	No leak / smooth operation
5	Re-instate PSV-3401 and all instruments before start-up	Record in equipment history

5. COMMON PROBLEMS & TROUBLESHOOTING

Symptom	Likely Cause	Action
Scheduled statutory PSV pop test	Set pressure drift +0.3 barg found	Overhauled PSV-3401, lapped seat, re-set to 6 barg, certified
End-of-life catalyst replacement at turnaround	Spent Pd/Al2O3 catalyst at end of run	Vacuumed out spent catalyst, inspected grids, loaded 3.2 m3 fresh cat.
Nitrogen leak at bottom manway during leak test	Spiral-wound gasket unevenly seated	Renewed SW SS316/graphite gasket, re-torqued in star pattern, leak tested.

6. KEY LEARNING POINTS

- A leaking manway loses inert gas and risks air ingress that could ignite the fresh, active catalyst.
- Respect the interlock SEQ-3401: e.g. TSHH-3401 trips at > 230 degC - never defeat it without a permit.
- Always cross-check the Datasheet limits and the P&ID TJC-LLD-PID-3401 before working on DC-3401A.
- Record the work in the equipment history so the Knowledge Hub stays the single source of truth.

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